



MIDSEMESTER



PHOTONIC DISCLINATION VORTEX CAVITY IN THz REGIME

Tera-QuaNTA Lab

Instrumentation and Applied Physics (IAP), IISc Bangalore

Presenter:

Shaukat Aziz

Sr no: 22171

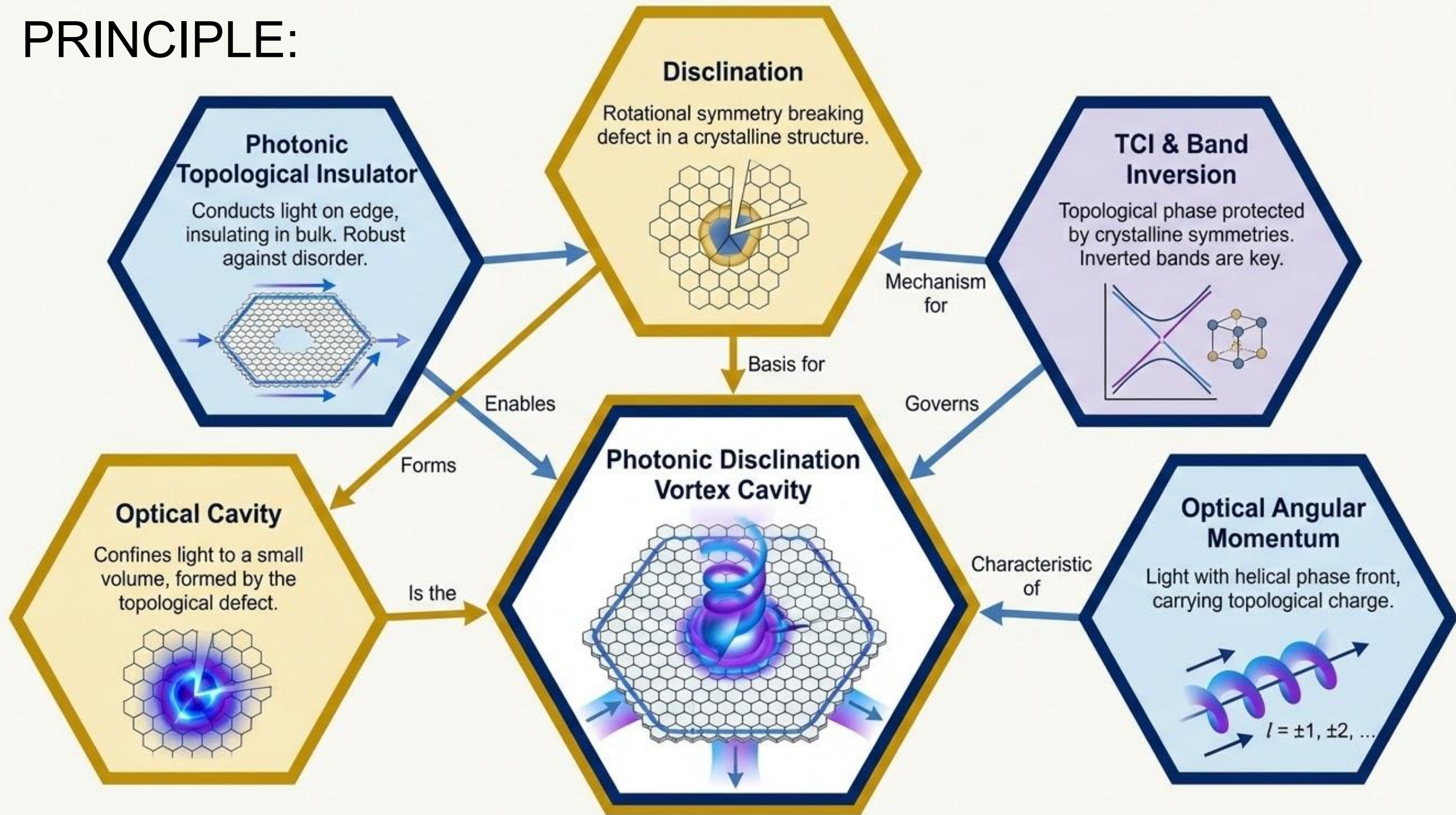
Bachelor of Science (Research) in Physics

Thesis Instructor:

Prof. Manukumara Manjappa



PRINCIPLE:

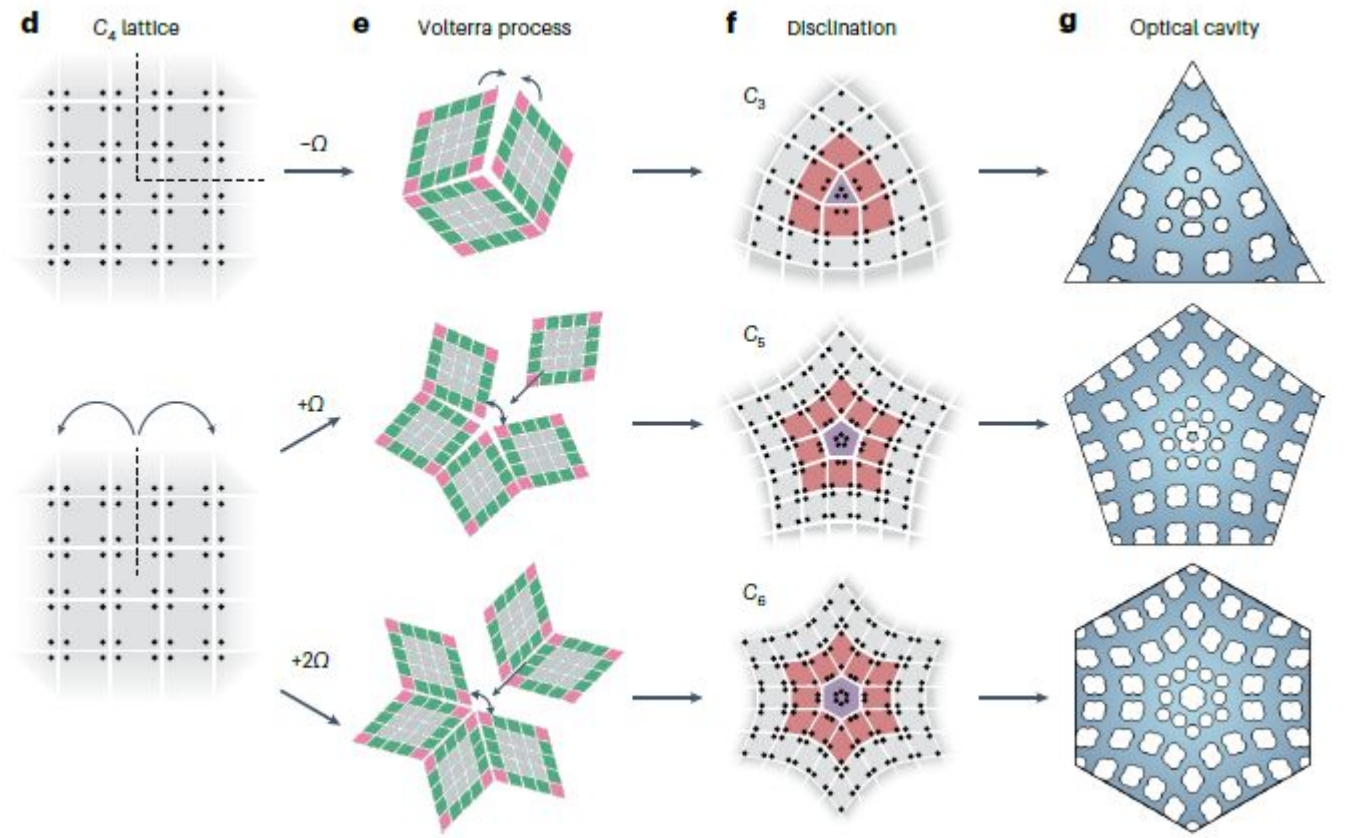


Cavity Architecture

Design Strategy

The **Volterra process** is a "cut-and-glue" surgery used to create topological defects in photonic crystals. A wedge-shaped sector of a symmetric lattice is removed (the **Frank angle**) Ω , and the remaining edges are joined. This breaks translational symmetry, forming a **disclination cavity** that traps light into high-quality vortex modes with specific topological charges. It is also radial invariance.

C4 to C3 removes 90° , stretching the remaining 270° to 360° . **C4 to C6** adds 180° , angularly compressing the total 540° into a flat 360° plane.



$$\Omega = (n - m) \cdot \frac{2\pi}{n} ,$$

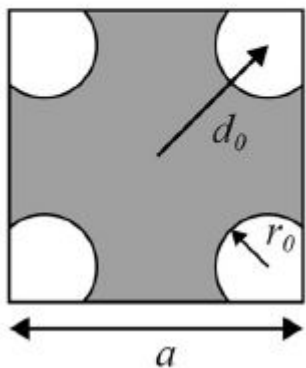
$$\theta' = \left(\frac{2\pi}{2\pi - \Omega} \right) \theta ,$$

$$C_4 \rightarrow C_3 : \quad \Omega = \frac{\pi}{2}, \quad \theta' = \frac{4}{3}\theta$$

$$C_4 \rightarrow C_6 : \quad \Omega = -\pi, \quad \theta' = \frac{4}{6}\theta$$

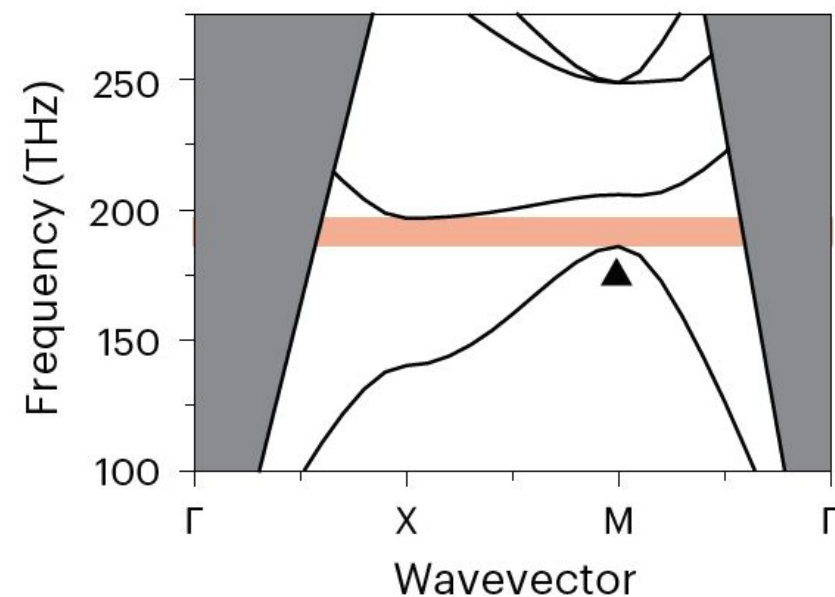
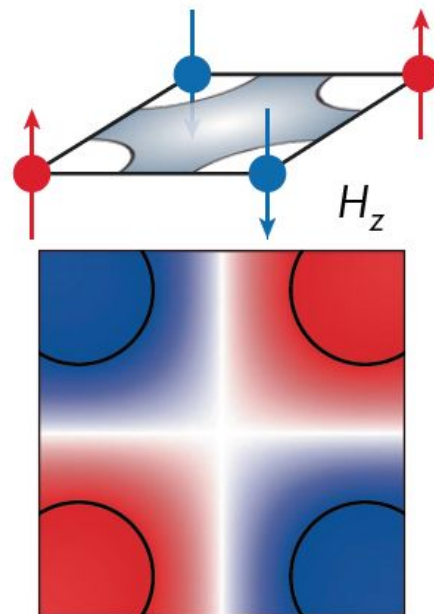
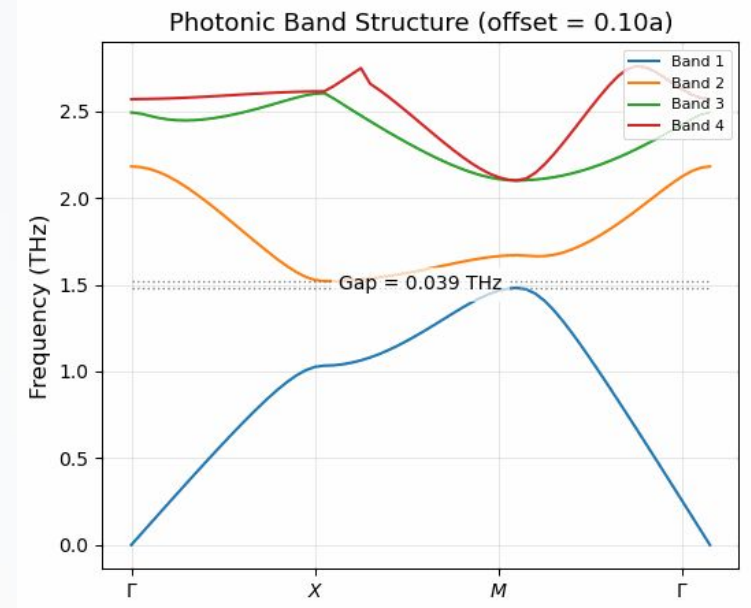
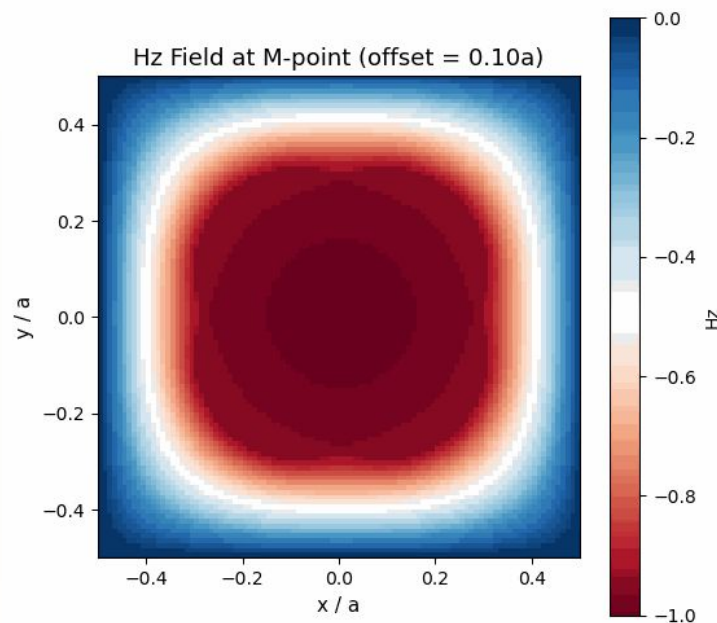
PLOTS

C_4 symmetric unit cell



$d_0 = \text{offset}$

Explanation:

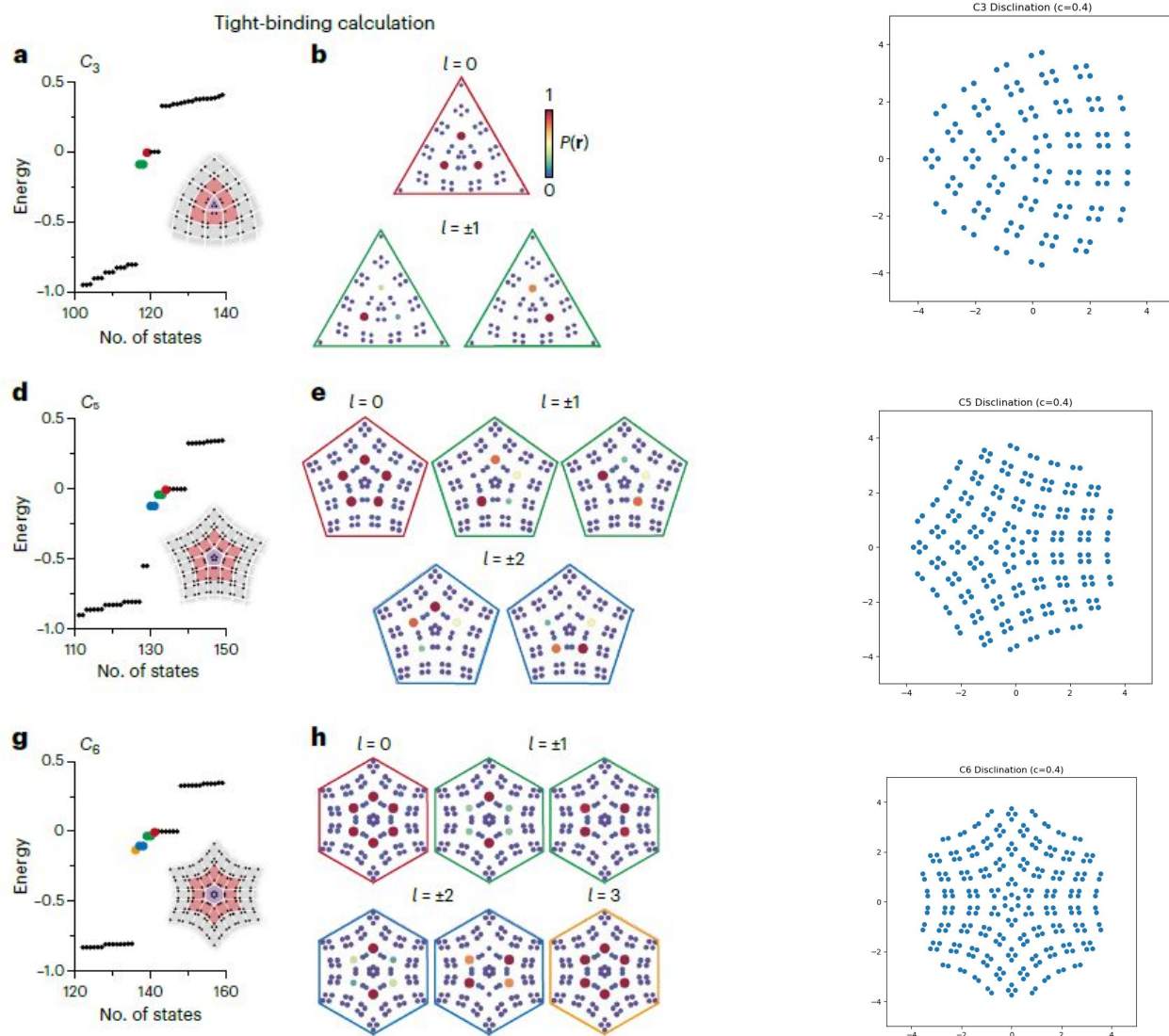




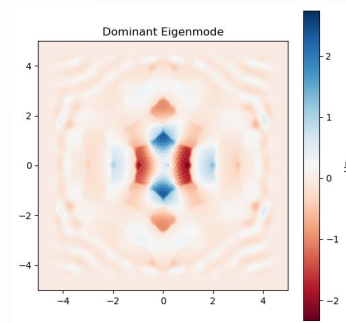
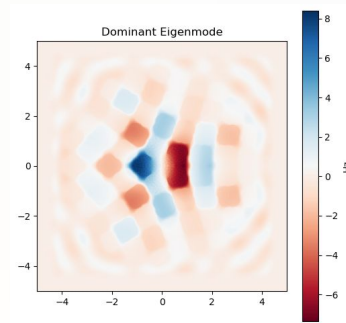
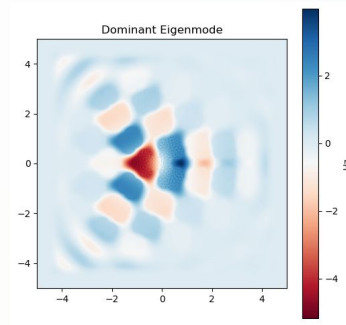
Work in Progress

Tight Binding and optical simulation of the cavity

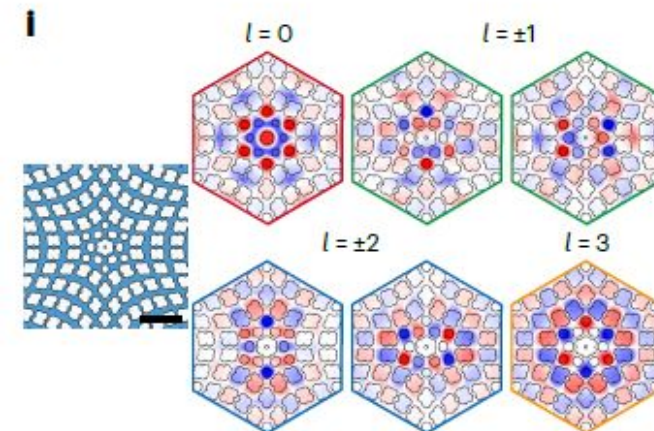
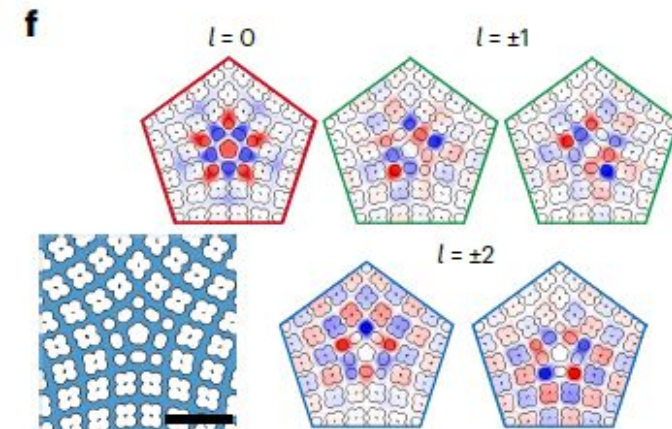
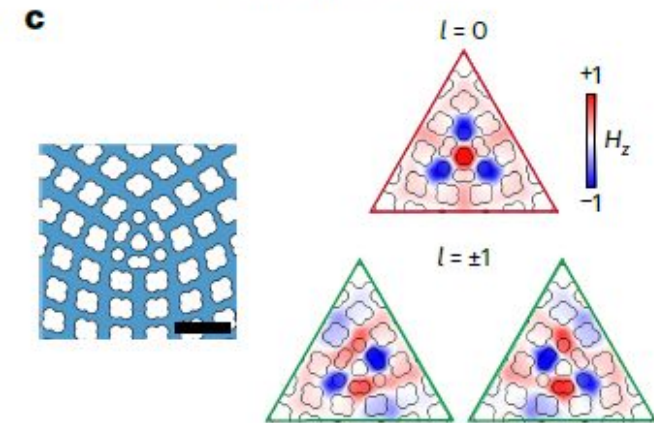
Topological Physics Foundations



The Terahertz (THz) Advantage



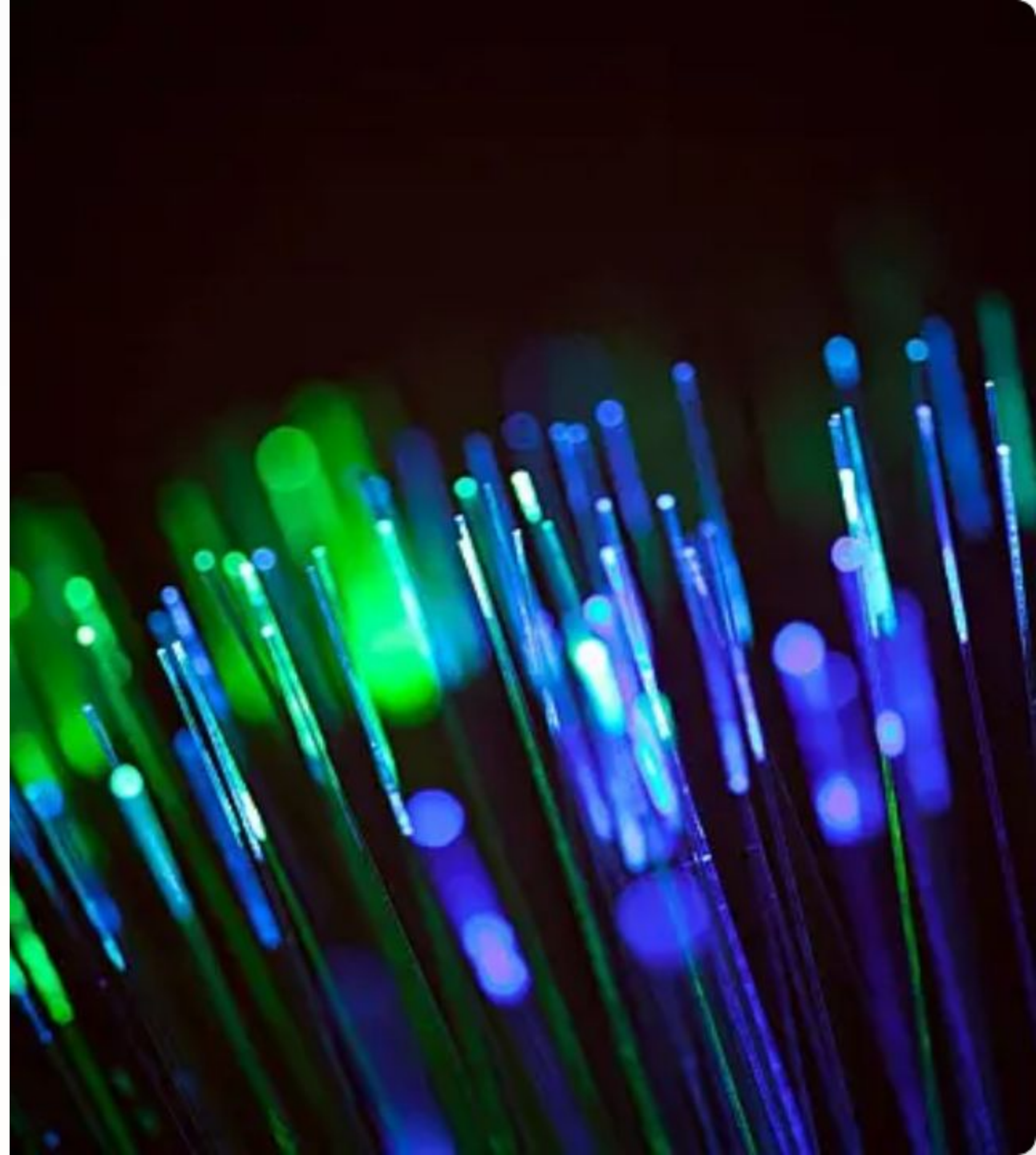
Optical simulation



Future Outlook

Topological Resilience

- Here we have just worked on c4 uniform lattice. This work can be further studied upon using Triangular lattice and other types of uniform lattice.
- This work paves the way for robust THz circuitry that is immune to structural defects, a major hurdle in current nano-fabrication.
- Future directions include integrating these cavities into active THz sensing as they are more sensitive.
- As this work was mainly for utilizing lasing originally, maybe feasibility of terahertz lasing can be tested.





Questions?

Thank you for your attention.

Presenter: Shaukat Aziz

Lab: Tera-QuANTA, IAP, IISc Bangalore

Email: shaukataziz@iisc.ac.in